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# Electrochemical Characterization of Single Lithium Ion Conducting Polymer Electrolytes Based on $sp^3$ Boron and Poly(ethylene glycol) Bridges

Gregorio Guzmán-González<sup>§,†,\*</sup>, Hugo J. Ávila-Paredes<sup>§</sup>, Ernesto Rivera<sup>‡</sup>, Ignacio González<sup>†\*</sup>

<sup>§</sup> Departamento de Ingeniería de Procesos e Hidráulica, Universidad Autónoma Metropolitana-Iztapalapa, Av. San Rafael Atlixco No. 186, México City, 09340, México

<sup>‡</sup> Instituto de Investigaciones en Materiales, Universidad Nacional Autónoma de México, Coyoacán 04510, México City, México

<sup>†</sup> Departamento de Química, Universidad Autónoma Metropolitana-Iztapalapa, Av. San Rafael Atlixco No. 186, 09340, Mexico City, Mexico

**KEYWORDS:** Lithium ion batteries; solid polymer electrolytes; boron  $sp^3$ ; transport number; non phase separation; single lithium ion conduction.

**ABSTRACT.** A novel series of single lithium ion conducting polymer electrolytes (SLICPE) based on  $sp^3$  boron and PEG bridges is presented, in the context of the development of a new generation of batteries, with the aim to overcome the problems related to concentration overpotential and low ion transport numbers in conventional solid polymer electrolytes (SPE).

The phase separation generated by the physical mixture of SPE with plasticizers such as poly(ethylene oxide) (PEO) is still a serious problem. In this work, the use of poly(ethylene glycol) (PEG) with different chain lengths, for the polycondensation reaction with  $LiB(OCH_3)_4$ , to synthesize SLICPE allows preventing phase separation, while tuning the predominant conduction mechanism, and thus the electrical properties, especially the lithium ion transference number. The ionic transport is promoted by chain mobility as the chain length is increased. SLICPE with the best ionic conductivity values  $(4.95 \pm 0.05) \times 10^{-6} \text{ S cm}^{-1}$  was the one synthesized from poly(ethylene glycol) with an average  $M_N$  of 400 (BEG8), having an  $O/Li^+$  ratio of 20. The lithium transference number ( $t_{Li^+}$ ) and electrochemical stability window of SLICPE membranes at 25 °C decreased as the PEG bridge length between  $sp^3$  boron atoms increased, from 0.97 to 0.88 and 5.4 V to 4.2 V vs  $Li^0/Li^+$ , respectively; for SLICPE synthesized from PEG with an average  $M_N$  of 50 to 400 (BEG 1 to BEG8).

## 1. Introduction

Lithium ion batteries, LIB, and lithium ion supercapacitors have dominated the market of energy storage systems for portable and mobile applications in the last few decades, due to their high gravimetric and volumetric energy densities.<sup>1</sup> In recent reviews of promising cathodic materials for LIB, lithium manganese nickel oxide (LMNO) and its derivatives, stand out among the materials with the highest redox potentials  $\sim 5$  V vs  $Li^0/Li^+$ .<sup>2-4</sup> However, the integration of materials with a high redox potential in commercial LIB has been hindered by the poor electrochemical stability of conventional liquid electrolytes based on lithium salts dissolved in organic carbonates (e.g. ethylene carbonate, EC; propylene carbonate, PC; dimethyl carbonate, DMC); such low stability causes the associated safety risks during the operation.<sup>5-8</sup>

Solid polymer electrolytes (SPE) have been proposed and widely studied<sup>9-10</sup> in order to substitute the commonly used liquid electrolytes, due to their low flammability, high thermal stability and wide electrochemical stability window. Such properties make SPE in most of the cases a safer alternative vs liquid electrolytes.<sup>11</sup> Classic SPE are formed by polymer matrices usually based on structures derived from poly(ethylene oxide) (PEO), and dissolved lithium salts (e.g.  $Li^+X^-$ : EO; X:  $PF_6^-$ ,  $TFSI^-$ ,  $CF_3SO_3^-$ ) of relatively small anions.<sup>12</sup> The ionic conductivity in those classic SPE is visualized as being due to a combination of an ion/polymer cooperative motion with occasional independent ion movements.<sup>13</sup> The anion does not interact significantly with the polymer chains but its motion requires free volume between the chains. A natural consequence of the structure of polymer electrolytes and the mechanism of ionic conduction is that anions tend to be more mobile than cations.<sup>14</sup> This is an indication of the dominant contribution of anions to the process of charge transference, which may be associated with the coupling of  $Li^+$  with the basic Lewis type sites in the polymeric matrix.<sup>7, 15</sup>

Relatively low ionic conductivity values (between  $10^{-5}$  and  $10^{-6}$  S cm<sup>-1</sup>) and lithium ion transference numbers ( $t_{Li^+}$ , generally less than 0.5), due to the simultaneous movement of anions and cations,<sup>16-17</sup> and the inferior interfacial properties of SPE are still key barriers for their application in large-scale batteries. Different methodologies have been successfully used to increase the ionic conductivity and  $t_{Li^+}$  and decrease the activation energy of ionic conduction processes in SPE, for instance: *a*) plasticizers, such as ionic liquids<sup>18-21</sup> and oligomers of low molecular weight (e.g. PEGME),<sup>22</sup> are able to increase the ionic mobility and the amorphousness in polymeric matrices; *b*) addition of micro and nano ceramic particles such as Al<sub>2</sub>O<sub>3</sub>, TiO<sub>2</sub> and SiO<sub>2</sub><sup>23-24</sup> (considered as inactive fillers) to SPE allows the generation of new conduction routes, so that ionic conductivity is increased, while interfacial secondary reactions are inhibited at the same time, thereby improving interfacial compatibility of the electrolyte with the electrodes<sup>25</sup>. However, the high surface energy of the above mentioned ceramic particles generates strong ceramic-polymer interactions that decrease the mobility of the polymer chains and substantially change the dominant mechanism for ionic conduction;<sup>26</sup> and *c*) the polymer electrolytes based on polymerized ionic liquids, PILs, or grafted oligomeric chains have been widely studied and proposed as strong candidates for use in LIB, due to their superior conductivity values.<sup>13</sup> Nevertheless, the concentration gradient generated by the accumulation of anions on the surface of the anode during the operation of LIB, with dual conducting electrolytes (whether liquids, gels or polymers) can give rise to secondary reactions, whose products increase the total impedance of the cells, thereby decreasing the overall performance of the LIB.<sup>11</sup>

The increase of  $t_{Li^+}$  in SPE can be achieved through the immobilization of anions into the main polymeric chains. This type of SPE, named single lithium ion conducting polymer electrolytes, SLICPE, exhibits  $t_{Li^+}$  values close to unity, making them significantly attractive for their use in

LIB,<sup>17, 27</sup> although their conductivities are lower than those of dual conducting polymer electrolytes.<sup>28-29</sup> SLICPE have been synthesized from precursors with charged delocalized anions of i) perfluoroether sulfonate-based,<sup>30-31</sup> ii) bis(sulfonyl) imide-based<sup>32</sup> or iii) boron atoms with an  $sp^3$  coordination<sup>29, 33</sup> among others.<sup>7, 34</sup> Synthesis of SLICPE based on boron atoms are among the simplest and with the lowest cost.<sup>35-36</sup>

Other boron compounds with a similar structure have been used as additives for both liquid electrolytes and SPE and include lithium borates, such as lithium bis(oxalate) borate (*LiBOB*) and particularly lithium oxalyldifluoro borate (*LiODFB*).<sup>37-38</sup> In these cases, the boron atoms are chelated with oxalates, which are used as electrolytic salts in LIB.<sup>39</sup> Furthermore, these compounds present a wide potential window of 4.5 to 6 V vs  $Li^0/Li^+$ , good thermal stability and form efficient solid electrolyte interfaces (SEI) during the stabilization of graphite anodes.<sup>40</sup> In addition, chelatoborates have been widely studied,<sup>41</sup> since the boron atoms have a  $p$ -orbital which can strongly interact with the basic anion of lithium salts, thereby causing the increase of  $t_{Li^+}$ , in both liquid electrolytes and SPE.<sup>36</sup>

The first reported SLICPE based on  $sp^3$  boron atoms was synthesized from the following precursors: lithium bis allyl malonate borate (*LiBAMB*) and an allyl group containing a comb-like branch, poly(pentaethylene glycol methyl ether acrylate co-allyl oxyethyl acrylate).<sup>42</sup> Such SLICPE exhibited conductivities in the order of  $10^{-8}$  S  $cm^{-1}$  at 30 °C under dry conditions and  $10^{-6}$  S  $cm^{-1}$  at 30 °C in the gel state with an EC/DMC (1:1, wt:wt ratio) solvent.<sup>22, 42</sup> Other synthesized SLICPE were lithium oxalate poly(acrylic acid borate) (*LiOPAAB*) and lithium borate oxalate poly(vinyl alcohol) (*LiPVAOB*),<sup>17</sup> based on poly(acrylic acid) (*PAA*) or poly(vinyl alcohol) (*PVA*), boric acid ( $H_3BO_3$ ), lithium hydroxide (*LiOH*) and oxalic acid ( $H_2C_2O_4$ ).<sup>43</sup> The ionic conductivities of these materials with PC as a solvent were up to  $10^{-6}$  S  $cm^{-1}$  at room

temperature, and the electrochemical stability window was up to 7.0 V vs  $Li^0/Li^+$ .<sup>22, 40</sup> In addition, it was demonstrated that the SLICPE with a higher  $Li^+$  content presented a higher level of rigidity, so that the conductivity followed an Arrhenius behavior. In this SLICPE the movement of  $Li^+$  then depends on the flexibility of the PEG polymer chains.

The mixture of lithium chelatoborates with low molecular weight poly(ethylene oxide) PEO, is one of the easiest ways to prepare SLICPE, modulating the mobility of the polymer chains through the concentration of chelatoborates. However, when they are mixed, a considerable phase separation is observed in the short term, thereby decreasing the performance and lifetime of the electrolytes.<sup>14, 44</sup>

On one hand, the formation of covalent bonds between ethoxy chains and  $sp^3$  boron atoms could avoid the problems generated by phase separation in SLICPE. Conversely, the use of short-range ethoxy chains could provide a suitable  $O/Li^+$  ratio that could favor the transport of  $Li^+$  through the polymeric matrix, increasing the ionic conductivity.

In this context, the present work shows the results of the electrochemical characterization of a series of SLICPE based on  $sp^3$  tetracoordinated boron atoms with poly(ethylene glycol) (PEG) chains with different molecular weights. The synthesis, based on a relatively simple and low cost synthetic approach, was performed by an alcoholic polycondensation, with lithium tetra methoxy borate ( $LiB(OCH_3)_4$ ), illustrated in Figure 1. The best  $O/Li^+$  ratio in SLICPE, that allows performing an enhancement of the electrochemical properties, was determined by the characterization of ionic conductivity,  $t_{Li^+}$  and electrochemical stability window by means of the electrochemical impedance spectroscopy (EIS), chronopotentiometry (CP) and linear sweep voltammetry (LSV), respectively.

## 2. Experimental

### 2.1. Reactants

Trimethyl borate (TMB), lithium metal ( $\text{Li}^0$ ), ethylene glycol (PEG50), di(ethylene glycol) (PEG100), tetra(ethylene glycol) (PEG200), poly(ethylene glycol) 300 (PEG300) with an average  $M_n = 300 \text{ g mol}^{-1}$ , poly(ethylene glycol) 400 (PEG400) with an average  $M_n = 400 \text{ g mol}^{-1}$  and methanol.

### 2.2. Synthesis of lithium tetramethoxy borate $\text{LiB}(\text{OCH}_3)_4$

Lithium tetramethoxy borate  $\text{LiB}(\text{OCH}_3)_4$  was synthesized and purified according to the procedure described by Barthel et al..<sup>41</sup> Lithium metal was mixed with methanol in a  $\sim 0.35 \text{ g} : 25 \text{ mL}$  ratio. Lithium was weighted and poured into a Pyrex flask inside a glove box; then, the flask was transferred from the glove box to a fume hood and immediately connected to an argon flow line. The flask was kept in an ice bath since the reaction is highly exothermic. Methanol was slowly added into the flask. Afterwards, the mixture was heated to  $60^\circ\text{C}$  to ensure that the lithium metal reacts completely with methanol. Then, trimethyl borate ( $5.2 \text{ g} ; 5.7 \text{ mL}$ ) was slowly dropped into the solution. At last, the reaction was kept for another 24 h at room temperature for crystallization. The sample was finally purified by desiccation. A white solid was obtained, with a yield of  $\sim 92 \%$ .

### 2.3. Synthesis of SLICPE samples

SLICPE based on boron atoms were synthesized, as shown in Figure 1, by the reaction of poly(ethylene glycol) (PEG) chains with different molecular weights and lithium tetramethoxy borate. ( $75 \text{ mg}, 5 \text{ mmol}$ ) of  $\text{LiB}(\text{OCH}_3)_4$  were mixed under constant magnetic stirring with 10



mmol of PEG (FW = 64 mg/mmol) in a 5 mL spherical flask under nitrogen atmosphere. The synthesis was carried out in two steps. In the first step, reagents were homogenized by magnetic stirring at 100 °C for 1 hour and in the second step, the polycondensation reaction was carried out at 180 °C for 20 min with a nitrogen flow in order to remove the methanol generated during the reaction and any traces of water from PEG reactants. In order to determine the polycondensation reaction temperature for the SLICPE synthesis, thermogravimetric analysis (TGA) of precursors was carried out (see below). No further purification methods were used, since the selected synthesis process does not involve the formation of by-products. Obtained samples, namely *BEG1*, *BEG2*, *BEG4*, *BEG6* and *BEG8* (the number in the samples nomenclature corresponds to that of the ethoxy groups between boron atoms), corresponding to precursors PEG50, PEG100, PEG200, PEG300 and PEG400, respectively, were stored in a glove box under argon atmosphere for later characterization.

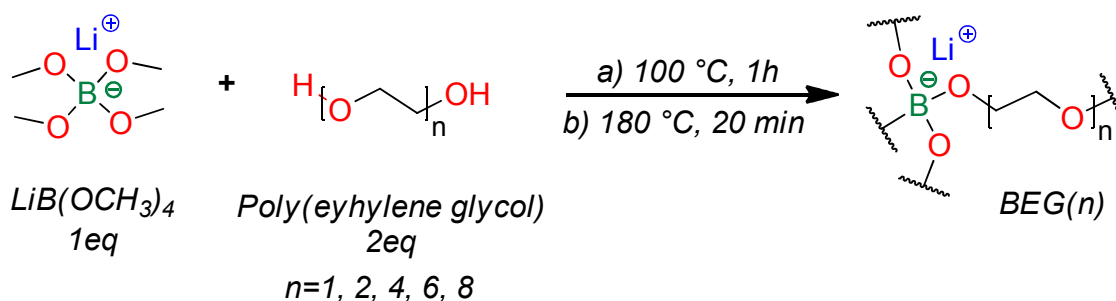


Figure 1. Scheme of the synthesis of SLICPE based on  $\text{sp}^3$  boron atoms: polycondensation of PEG chains with different molecular weights.

## 2.4. Characterization of SLICPE

$^1\text{H}$  and  $^{13}\text{C}$  NMR spectra were recorded in a Bruker AVANCE-III DMX500 NMR using  $\text{D}_2\text{O}$  as solvent and trimethylsilylpropanoic acid (TSP) as reference. Solid state  $^{11}\text{B}$  and  $^7\text{Li}$  NMR spectra were acquired at room temperature, on MAS conditions at (6 kHz) and HPDEC technique, respectively, in a Bruker AVANCE-II 300 NMR spectrometer. Boric acid was used as reference of the  $^{11}\text{B}$  chemical shift (chemical shifts are reported in ppm). Attenuated total reflection Fourier transform infrared (ATR-FTIR) spectra of the samples were obtained in the range from 4000 to  $550\text{ cm}^{-1}$ , using a Nicolet (model 6700) spectrometer, based diamond and ATR accessory (model Smart Orbit); 10 scans were averaged with a resolution of  $2\text{ cm}^{-1}$ .

Thermogravimetric analyses (TGA) were performed in a TGA Q5000 Instrument with a nitrogen flow of  $50\text{ cm}^3\text{ min}^{-1}$  in the range from 30 to  $700\text{ }^\circ\text{C}$  at  $10\text{ }^\circ\text{C min}^{-1}$ . Differential scanning calorimetry (DSC) was carried out in a DSC Q2000 instrument, with a nitrogen flow of  $50\text{ cm}^3\text{ min}^{-1}$ , at a heating rate of  $10\text{ }^\circ\text{C min}^{-1}$  (first scan), to determine the melting point  $T_m$  and glass transition temperature  $T_g$  of the samples, an amplitude of  $\pm 1.06\text{ }^\circ\text{C}$  and a period of 40 s in the temperature range from  $-70$  to  $150\text{ }^\circ\text{C}$ . X-ray diffraction (XRD) patterns of the samples were obtained in an advanced x-ray diffractometer (Bruker D-8 with geometry Bragg-Brentano), using a  $\text{Cu K}\alpha$  radiation, with a scanning rate of  $1^\circ\text{ min}^{-1}$  in the  $2\theta$  range from 2 to  $70^\circ$ .

## 2.5. Electrochemical measurements of SLICPE

**Ionic conductivity** ( $\sigma_{DC}$ ) was estimated from electrochemical impedance spectroscopy (EIS) measurements. A symmetrical stainless steel/polymer electrolyte/stainless steel cell was assembled. The distance between the electrodes (L) was kept  $\sim 0.1\text{ cm}$ , using a teflon spacer ring with an inner area (A) of  $0.24\text{ cm}^2$ . EIS measurements were performed applying a 10 mV amplitude perturbation in the frequency range from 1 MHz to 10 mHz at open circuit potential

(OCV) conditions. The ohmic resistance ( $R_0$ ) of the sample, estimated from the Nyquist plot at the low frequency end of the semicircle, was used to calculate the ionic conductivity using the following equation:

$$\sigma = \frac{L}{(A R_0)}, \text{ (S cm}^{-1}\text{)} \quad (1)$$

Measurements were carried out from 20 to 90 °C at every 10 °C interval; the temperature was controlled, using a circulator coupled to a temperature bath (SEV, FC-10). Cells were allowed reaching thermal equilibrium for at least 1.5 h for their first test, and 30 min for the following tests.

**The lithium-ion transference number ( $t_{Li^+}$ )** was determined by the AC-DC polarization experiment according to the Evans-Bruce protocol (Eq. (2)).<sup>45</sup> Symmetrical Swagelock cells,  $Li^0/SLICPE/Li^0$ , were assembled (SLICPE films of ~ 0.1 cm of thickness and 1.0 cm diameter) and subjected to a 10 mV polarization bias ( $\Delta V$ ) to determine the initial ( $I_0$ ) and steady state ( $I_s$ ) currents. EIS was performed applying a 10 mV perturbation, in the frequency range from 1 MHz to 10 Hz, at open circuit conditions to obtain the resistance of the passivation layer before ( $R_0$ ) and after ( $R_s$ ) polarization. Lithium ion transference number ( $t_{Li^+}$ ) was calculated using the following equation:

$$t_{Li^+} = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)} \quad (2)$$

**Linear sweep voltammetry** for the as-prepared SILCPE was carried out in a two-electrode Swagelock stainless steel/SLICPE/ $Li^0$  cell. Stainless steel was used as the working electrode and lithium foil was used as the counter and reference electrode. The measurements were performed from 0 to 6 V vs.  $Li^0/Li^+$  at a scan rate of 10 mV s<sup>-1</sup> at 30 °C.

**Compatibility evaluation of the proposed materials with a Li metal anode.** To perform the evaluation of the impedance after several days, symmetrical *ECC-Combi* cells,  $\text{Li}^0/\text{SLICPE}/\text{Li}^0$  were assembled with SLICPE films of  $\sim 0.05$  cm of thickness and 0.6 cm diameter.

All the electrochemical techniques described above were performed on a Multi-Potentiostat/Galvanostat VMP3 from Bio-Logic Science Instruments using a conductivity cell,  $\text{LiB}(\text{OCH}_3)_4$  swagelok cell and *ECC-Combi* used as ECC-Std electrochemical test cell for two-electrode testing. The cells were assembled inside an argon-filled glovebox (MBraun UNILab,  $\text{H}_2\text{O}$  and  $\text{O}_2$  contents  $< 0.5$  ppm) to avoid water adsorption and heated to  $60^\circ\text{C}$  in a BINDER oven for at least 24 h prior to any measurement.

### 3. Results and Discussion

The formation of borate groups in  $\text{LiB}(\text{OCH}_3)_4$  was corroborated by NMR spectroscopy, while efficiency of the SLICPE synthesis process by the polycondensation method was evaluated by solid state NMR, determining the  $\text{BO}_3 / \text{BO}_4^-$  ratio in SLICPE as a function of the length of the PEG chain. Likewise, the mobility of the polymer chains was evaluated by determining the glass transition temperature and the effect of this mobility on the crystallinity of the different synthesized SLICPE. In order to determine the ionic conducting properties, electrochemical characterization techniques were used for elucidation of the relations of the PEG chain lengths of SLICPE with the controlling  $\text{Li}^+$  conduction mechanisms, and activation energies of the ionic conduction processes.

### 3.1 NMR

The chemical structures of SLICPE prepared by polycondensation were confirmed by  $^1\text{H}$  and  $^{13}\text{C}$  NMR spectroscopies. Results are shown in Figure 2. The  $^1\text{H}$  chemical shift located between 3.65 and 3.7 ppm was assigned to the protons of  $\sim\text{BO}-\text{CH}_2-\text{CH}_2\sim$  for *BEG1* (Figure 2*ai*). The  $^1\text{H}$  chemical shifts at 3.65 and 3.75 ppm, in Figure 2*aii*, were assigned to two triplets of protons alpha and beta of the  $\sim\text{BO}-\text{CH}_2-\text{CH}_2-\text{OC}\sim$  for *BEG2* (Figure 2*a:α,β*). Another signal with a chemical shift of 3.7 ppm appears in the spectra of *BEG4*, *BEG6* and *BEG8*, which is assigned to PEG ether protons of  $\sim\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_2\sim$  (Figure 2*a:γ*). The intensity of this signal augments as the number of alpha and beta protons increase (due to the increase of the PEG chains length; Figure 2*aiii-v*). A small signal associated with the methoxy groups of residual  $\text{LiB}(\text{OCH}_3)_4$  is observed at 3.34 ppm (Figure 2*ai-ii*). The absence of the signal due the terminal -OH groups of PEG, which appears at ca. 2.9 ppm, confirms that the polycondensation successfully occurred. In the  $^{13}\text{C}$  NMR spectra (Figure 2*b*), the  $^{13}\text{C}$  chemical shifts that appeared at (60.36 and 60.50), 71.5, and 69.58 ppm are assigned to ( $\alpha$ )  $\sim\text{BO}-\text{C}-\text{C}-\text{OB}\sim$ , ( $\beta$ )  $\sim\text{BO}-\text{C}-\text{C}-\text{OC}\sim$  and ( $\gamma$ )  $\sim\text{CO}-\text{C}-\text{C}-\text{OC}\sim$  carbons, respectively.<sup>4</sup>

The borate structures of the SLICPE were investigated by solid-state CP/MAS  $^{11}\text{B}$  and  $^7\text{Li}$  NMR (Figure 3). In amorphous and crystalline borate compounds, boron may exist in tetrahedral  $\text{BO}_4^-$  and trigonal  $\text{BO}_3$  units.<sup>46-48</sup>  $^{11}\text{B}$  NMR spectra of tetrahedral  $\text{BO}_4^-$  shows a relatively narrow single strong signal in the range between 7 and 9 ppm, like the signal of the reference ( $\text{LiB}(\text{OCH}_3)_4$ ) (Figure 3*ai*), which indicates a highly symmetrical arrangement of the four oxygens in the  $\text{BO}_4^-$  tetrahedron in the chemical structure, mainly when the PEG chains are long (Figure 3*av-vi*).

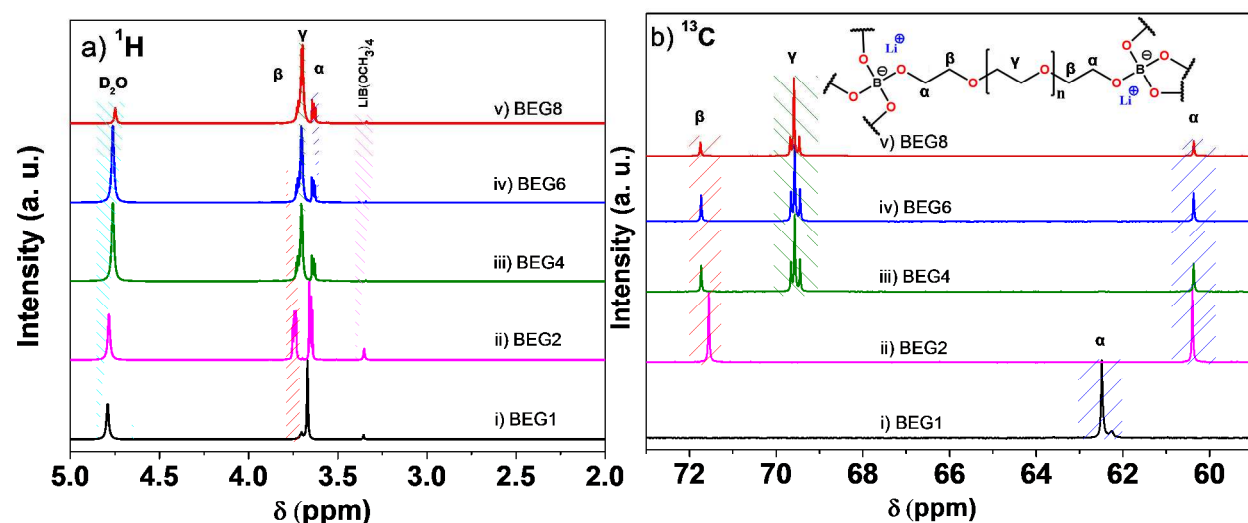


Figure 2. a)  $^1\text{H}$  and b)  $^{13}\text{C}$  NMR spectra in  $\text{D}_2\text{O}$  for synthesized boron-based SLICPE, with different PEG bridges length; i) *BEG1*, ii) *BEG2*, iii) *BEG4*, iv) *BEG6* and v) *BEG8*. The chemical groups associated to the signals are indicated in the figure ( $\alpha$ )  $\sim\text{BO}-\text{CH}_2-\text{CH}_2\sim$ , ( $\beta$ )  $\sim\text{BO}-\text{CH}_2-\text{CH}_2-\text{OC}\sim$  and ( $\gamma$ )  $\sim\text{CH}_2-\text{O}-\text{CH}_2-\text{CH}_2\sim$ .

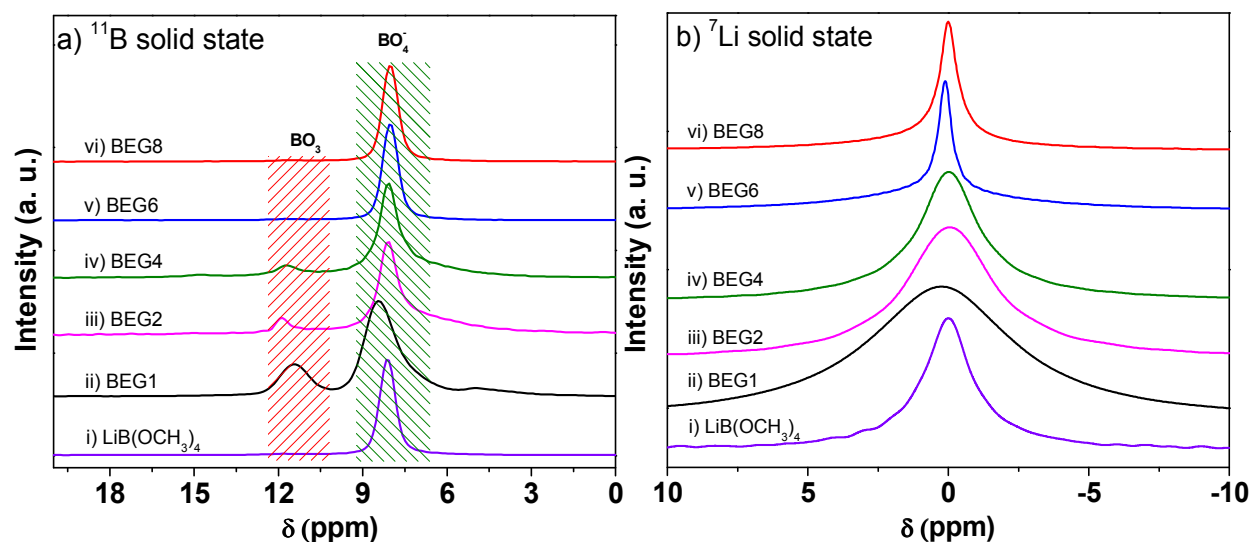


Figure 3. a)  $^{11}\text{B}$  and b)  $^7\text{Li}$  NMR/MAS spectra of synthesized boron-based SLICPE, with different PEG bridges length; i)  $\text{LiB}(\text{OCH}_3)_4$ , ii) *BEG1*, iii) *BEG2*, iv) *BEG4*, v) *BEG6* and vi) *BEG8*. The boron chemical species associated to each signal are indicated in the figure.

Generally, the chemical shift range of  $^{11}\text{B}$  is small and the peaks corresponding to the two boron coordinates cannot often be clearly separated. However, the distance increase between boron atoms diminishes the interactions  $^{11}\text{B}$ - $^{11}\text{B}$ ,  $^{11}\text{B}$ - $^{10}\text{B}$ , while  $^{11}\text{B}$ - $^1\text{H}$  interactions are eliminated by intense proton decoupling. These facts allow the rapid acquisition of  $^{11}\text{B}$  NMR spectra of the borates, from the determination of accurate trigonal  $\text{BO}_3$  / tetrahedral  $\text{BO}_4^-$  ratio.<sup>49</sup> The use of low molecular weight PEG can turn away the polycondensation reaction towards the formation of different polycrystalline boron compounds with thermodynamically more stable molecular structures,<sup>50</sup> typically  $\text{sp}^2$  boron atoms. In this sense, the highest value of the  $(\text{BO}_3 / \text{BO}_4^-)$  ratio is obtained for the sample *BEG1* (Figure 3a $_{ii}$ ). The use of higher molecular weight PEG favours the formation of  $\text{BO}_4^-$  groups, thereby decreasing the  $(\text{BO}_3 / \text{BO}_4^-)$  ratio, conserving the total boron atoms with a  $\text{sp}^3$  hybridization, for *BEG8* SLICPE (Figure 3a $_{vi}$ ).

The CP/MAS  $^7\text{Li}$ -NMR spectra of synthesized SLICPE are shown in Figure 3b. The spectrum of *BEG1* with a  $\text{O}/\text{Li}^+$  ratio of  $\sim 4$  exhibited a wide  $^7\text{Li}$  resonance at 0.2 ppm (Figure 3b $_{ii}$ ), which is characteristic of  $\text{Li}^+$  in concentrated polymeric electrolytes and vitreous states with a relatively static environment.<sup>50</sup> The spectra of samples *BEG2* and *BEG4* with  $\text{O}/\text{Li}^+$  ratios of  $\sim 6$  and 10, respectively, exhibited a wide  $^7\text{Li}$  resonance at 0.06 ppm, characteristic of  $\text{LiB}(\text{OCH}_3)_4$  salt (Figure 3b $_{iii-iv}$ ). Such a compound would increase the mobility of  $\text{Li}^+$  limited by the rigidity of the polymeric matrix.

On the other hand,  $^7\text{Li}$  NMR spectra of samples *BEG6* and *BEG8*, with  $\text{O}/\text{Li}^+$  ratios of  $\sim 14$  and 20, respectively, show narrow resonance signals at 0.1 and 0.06 ppm, characteristic of high mobility of  $\text{Li}^+$ . The differences of these two last spectra with those already described are due to the increase in the amount of dissociated  $\text{Li}^+$  (freely mobile for hopping) as the PEG chain

length increased; since more dissociation would lead to fewer inter-polymer interactions, increasing the mobility of  $Li^+$  in the electrolytes (*See below*). The above mentioned NMR characterization confirms well the expected chemical structures of the synthesized SLICPE. However, PEG chains between  $sp^3$  boron atoms give rise to different structural arrangements depending on the PEG chain length. These arrangements were characterized by XRD to determine the minimum chain length to obtain an amorphous material capable of having high free volume and thus, a high PEG chain mobility that enhances the  $Li^+$  transport.

### 3.2. XRD

XRD pattern of  $LiB(OCH_3)_4$  shows high intense peaks at angles  $2\theta = 17.4, 19.1, 20.4, 21.4, 25.9$  and  $35.2^\circ$ , which reveals the crystalline structure of the salt (Figure 4i). The XRD patterns of SLICPE present a broad peak, which can be observed around  $2\theta = 22.2^\circ$  that is characteristic of amorphous polymers.<sup>47, 51</sup>

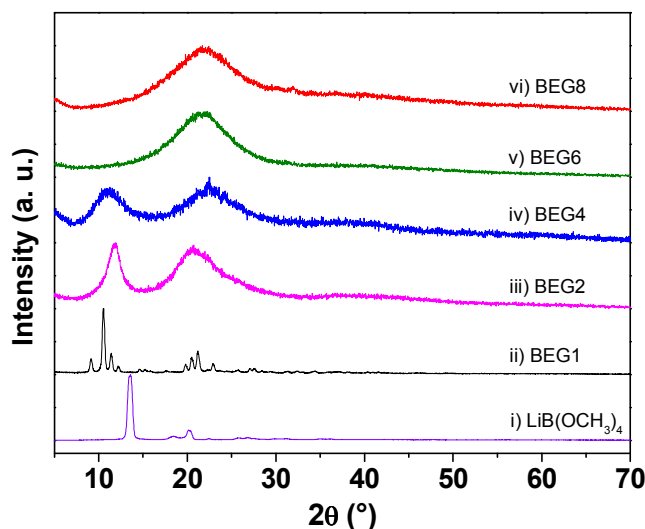


Figure 4. XRD patterns of synthesized boron-based SLICPE, with different PEG bridges length; i)  $LiB(OCH_3)_4$ , ii)  $BEG1$ , iii)  $BEG2$ , iv)  $BEG4$ , v)  $BEG6$  and vi)  $BEG8$ .



1  
2  
3 In addition to the above mentioned broad peak at  $2\theta = 22.2^\circ$ , the XRD pattern of *BEG1* shows a  
4  
5 peak at  $2\theta = 10.6^\circ$  (Figure 4ii), which suggests the formation of secondary crystalline phases  
6  
7 such as boroxine rings or glasses with different boron/oxygen stoichiometry ratios, derived from  
8  
9 the polycondensation reaction between PEG50 and  $\text{LiB}(\text{OCH}_3)_4$ .<sup>52</sup>  
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12  
13 PEG itself is an amorphous polymer which does not show any melting point. However, after the  
14  
15 modification with  $\text{LiB}(\text{OCH}_3)_4$ , the presence of anionic borates with their corresponding  $\text{Li}^+$   
16  
17 counter-ions (ionic bond) give rise to the formation of crystalline domains. This can be  
18  
19 confirmed by the presence of melting points in the DSC of all the samples (Table 1). When the  
20  
21  $\text{O}/\text{Li}^+$  ratios are small  $\text{Li}^+$  mainly interacts with  $\text{BO}_4^-$ , since the borate anions have a higher  
22  
23 electronegativity than the oxygen atoms of the PEG chains. Moreover, the XRD patterns of  
24  
25 *BEG2* and *BEG4* exhibited a broad peak between  $2\theta = 11.9^\circ$  and  $2\theta = 11.3^\circ$  (Figure 4iii-iv),  
26  
27 which indicates the formation of crystalline domains (i. e. different to the crystalline phase  
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29 mentioned for sample *BEG1*; Figure 4ii).  
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35 However, in the samples *BEG6* and *BEG8* with  $\text{O}/\text{Li}^+$  molar ratios of 14 and 20 respectively,  
36  
37 the  $\text{O}/\text{Li}^+$  ratio is low enough to achieve total dissociation of  $\text{Li}^+\text{-BO}_4^-$  complexes with the  
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39 oxygen atoms of the PEG chains. The decrease of the interaction strength between  $\text{Li}^+$  and boron  
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41 atoms gives rise to completely amorphous phases formation, characterized by a single wide peak  
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43 at  $22.2^\circ$  in XRD patterns (Figure 4v-vi).  
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48 The XRD technique allowed the elucidation of the loss of crystallinity in the SLICPE, attributed  
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50 to the increase in the PEG chains length between  $sp^3$  boron atoms and the breaking of  $\text{Li}^+$  and  
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52  $\text{BO}_4^-$  complexes by the increase in mobility of the polymer chains. However, more specific  
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information about the mobility of the polymeric chains in SLICPE is obtained through the characterization by DSC.

3.3. Thermogravimetric analysis (TGA)

TGA of stoichiometric mixtures of precursors ( $LiB(OCH_3)_4$  and PEG of different  $M_N$ ) were performed to determine the synthesis temperature as well as to evaluate the thermal stability of SLICPE. Figure 5 shows the weight loss curves of the mixtures of precursors for SLICPE BEG1 and BEG8 (corresponding to the shortest and the longest PEG chains, respectively). As indicated in the figure, reaction temperature is located in the  $\sim 180$ - $200$  °C range, but a sharp weight loss starts after  $\sim 380$  °C and  $\sim 255$  °C, respectively for BEG1 and BEG 8, so that these values could be considered the temperature limits for stability; the indicated weight loss percentage values correspond to the loss of methanol (byproduct of the polycondensation reaction). The stability of SLICPE decreases as the PEG length increases, where  $255$  °C is the lowest value (for BEG8).

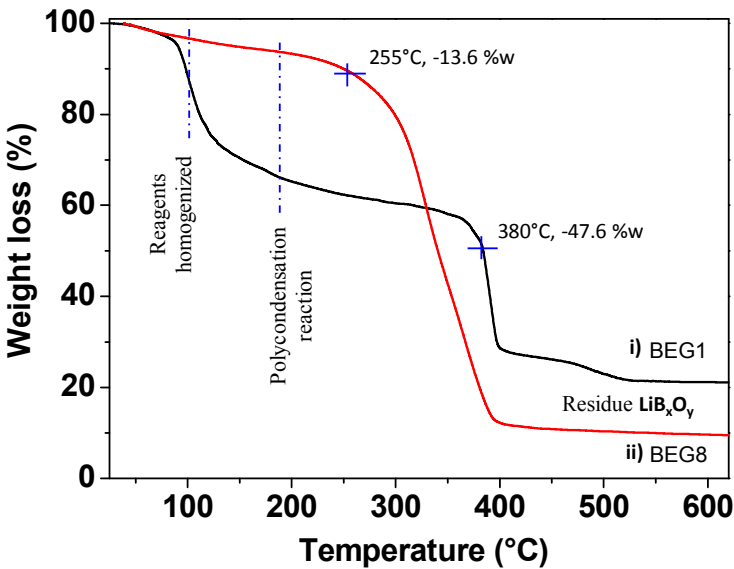


Figure 5. Weight loss curves of the mixtures of precursors for SLICPE, with different PEG bridges length; i) *BEG1* and ii) *BEG8*. The different steps of mixture transformation are indicated in the figure.

### 3.4. Differential scanning calorimetry (DSC)

Differential scanning calorimetry was used to determine the glass transition temperature ( $T_g$ ) and melting temperature ( $T_m$ ) of synthesized SLICPE (Figure 6). A summary of the  $T_g$  and  $T_m$  values is shown in Table 1. The  $T_g$  of samples decreased from 15 to  $-42 \pm 2^\circ\text{C}$  when the  $O/\text{Li}^+$  molar ratio varies from 4 to 20 for *BEG1* to *BEG8*, respectively, which is reasonable since pure PEG shows an average  $T_g$  of  $-67^\circ\text{C}$ . The presence of boride with the cross-linking effect restrains the mobility of the polymer chains thereby increasing the  $T_g$  of the system with respect to PEG. There is also a decrease in rigidity of the polymeric structure when the PEG chains length between the  $sp^3$  boron atoms is longer. Increasing the free volume, the movement of the polymeric matrix is facilitated and consequently, the ionic transport could be promoted in *BEG6* and *BEG8*.<sup>19, 47</sup>

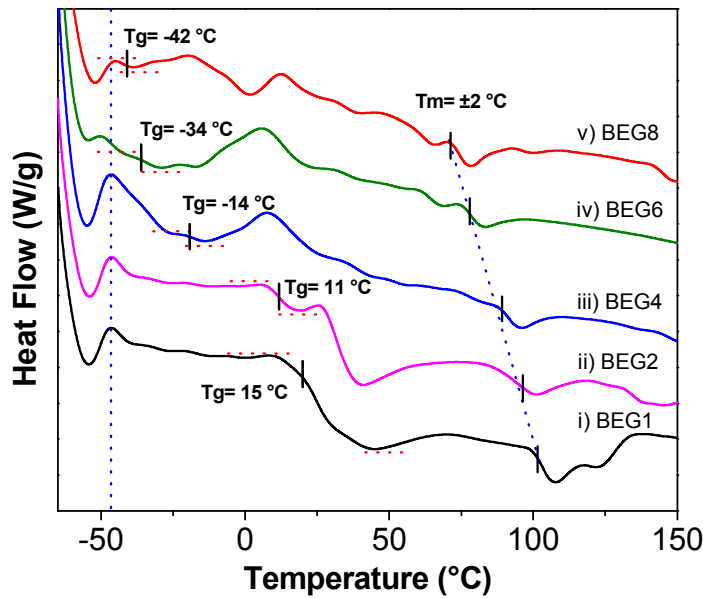


Figure 6. Differential scanning calorimetry (DSC) of synthesized boron-based SLICPE, with different PEG bridges length; i) *BEG1*, ii) *BEG2*, iii) *BEG4*, iv) *BEG6* and v) *BEG8*. The temperature region used for *Tg* evaluation is indicated in the figure

Sample *BEG1* exhibited the highest average *Tm* value (99 °C; Figure 6i), which has the contribution of the melting temperature of semi-crystalline regions and the complete mobility state of the polymer chains. Besides *Tm*, thermal parameters including melting enthalpy during heating ( $\Delta H_m$ ), and relative crystallinity (*Xc*) were obtained and are shown in Table 1. *Xc* is calculated by comparing  $\Delta H_m$  to the theoretical value of 100 % crystallized crystal ( $\Delta H_m^0$ ), taken as reference  $\Delta H_{m(PEO)}^0 = 213.17 \text{ J/g}$ ;<sup>53</sup> and  $Xc = \Delta H_m / \Delta H_{m(PEO)}^0 \times 100 \%$ . The decrease of  $\Delta H_m$  and *Tm* of SLICPE with the increasing PEG chain length reveals that relative crystallinity *Xc* diminishes from 7.5 to 1.6 % for BEG1 and BEG8, respectively during the limited time of measurement. Generally, *Tm*, *Tg* and *Xc* in SLICPE decrease as the PEG chain length (between *sp*<sup>3</sup> boron atoms) increases.<sup>54</sup>

TABLE 1. Thermal properties obtained by DSC from experimental data shown in Fig. 6.

| SLICPE      | molar ratio       |                  | % Crystallinity ( $X_c$ ) |                                    |
|-------------|-------------------|------------------|---------------------------|------------------------------------|
|             | O/Li <sup>+</sup> | $T_g \pm 2$ (°C) | $T_m \pm 2$ (°C)          | $\Delta H_m / \Delta H_{m(PEO)}^0$ |
| <i>BEG1</i> | 4                 | 15               | 99                        | 7.5                                |
| <i>BEG2</i> | 6                 | 11               | 98                        | 5.9                                |
| <i>BEG4</i> | 10                | -14              | 88                        | 2.9                                |
| <i>BEG6</i> | 14                | -34              | 82                        | 1.7                                |
| <i>BEG8</i> | 20                | -42              | 78                        | 1.6                                |

This behavior is related to an enhance in the polymer chains mobility, thereby suggesting an increase in total ionic conductivity. However, the decrease of  $Li^+$  concentration by the increase in the PEG chains length between boron atoms can affect the ionic conductivity in SLICPE.<sup>24, 55</sup> In this sense, electrochemical impedance spectroscopy (EIS) is used to quantify the conductivity and obtain relevant information about the predominant ionic conduction mechanisms in SLICPE (Figure 7).

### 3.4. Electrochemical measurements of SLICPE

From synthesized SLICPE, samples *BEG1* and *BEG2* exhibited the lowest ionic conductivities (i. e.  $(4.37 \pm 0.89) \times 10^{-10}$  S cm<sup>-1</sup> and  $(9.97 \pm 0.03) \times 10^{-9}$  S cm<sup>-1</sup>, respectively, at 30 °C), even though the O/Li<sup>+</sup> ratio is the lowest, implying that those samples have the highest limitations for ionic mobility. However, these limitations in the set of synthesized samples were easily overcome by

increasing the PEG chains length. *BEG4*, *BEG6* and *BEG8* exhibited ionic conductivities of  $(6 \pm 0.04) \times 10^{-7}$ ,  $(2.65 \pm 0.04) \times 10^{-6}$  and  $(4.95 \pm 0.02) \times 10^{-6} \text{ S cm}^{-1}$  at 30 °C, respectively (Figure 7). The tendency of ionic conductivity values obtained at room temperature indicated that ionic mobility in the synthesized SLICPE is mainly governed by the mobility of the PEG segments between  $sp^3$  boron atoms.

The temperature dependence of ionic conductivity determined by EIS is shown in Figure 7. The thermally activated behavior of conductivity is often described by the Arrhenius (Eq. (3)) or the Vogel-Tamman-Fulcher (VTF) equations (Eq. (4)), depending on the relationship between ionic movement and mobility of the polymer chains, determined by  $T_g$ ,

$$\sigma = AT^{-1} \exp\left(\frac{-E_A}{k_B T}\right) \quad (3)$$

$$\sigma = A^*T^{-1/2} \exp\left(\frac{-B}{k_B(T-T_0)}\right) \quad (4)$$

where  $A$  and  $A^*$  are the pre-exponential factors of the conductivity defined by Eq.(3) and (4), respectively (constants which are proportional to the concentration of mobile ions);  $B$  denotes the pseudo-activation energy associated with the motion of the polymer segment;  $k_B$  represents the Boltzmann constant;  $E_A$ , the activation energy, and  $T_0$  is a reference temperature (normally associated with the ideal  $T_g$  at which the free volume is zero, or with the temperature at which the configurational entropy becomes zero).<sup>56</sup> Since  $T_0$  is difficult to obtain experimentally, in this work it was considered as 50 K below the glass transition temperature,  $T_g$ , determined by differential scanning calorimetry (DSC).<sup>57</sup>

The low linearity of the curves in Figure 7a suggested that ionic conductivity of synthesized SLICPE does not follow an Arrhenius behavior. A better correlation of experimental data was

obtained by fitting with the Vogel-Tamman-Fulcher (VTF) equation (Figure 7b), indicating that ionic conduction strongly depends on the PEG chains movement. The values of parameters  $A^*$ ,  $B$  and  $T_0$ , obtained from fitting of experimental data, are shown in Table 2.

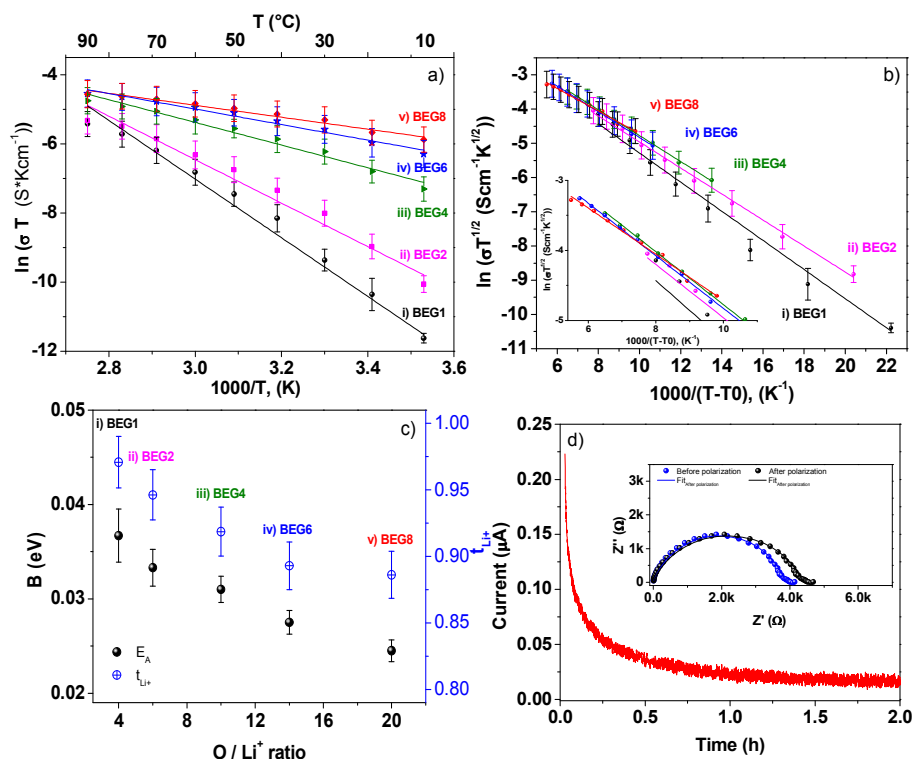


Figure 7. Ionic conductivity of the synthesized boron-based SLICPE, with different PEG bridges length; **a)** Arrhenius plots **b)** VTF plots, **c)** Variation of pseudo-activation energy  $B$  (evaluated from analysis of fig 7b) and  $t_{Li^+}$  (evaluated from AC/DC method, Eq. (2)) with the  $O/Li^+$  ratio of SLICPE and **d)** Lithium transfer number ( $t_{Li^+}$ ) evaluation: Typical Current transient obtained at polarization of 10 mV for Li/BEG4/Li cells at 25 °C (inset: Nyquist plot for BEG4 for the same cell before and after polarization).

Table 2. Influence of  $O/Li^+$  ratio on the ionic conductivity and transport number for SLICPE. The parameter values obtained from the fitting of experimental data shown in Figure 7b with Vogel–Tamman–Fulcher equation (4) are included.

| Sample | ratio                  | $\sigma_{25^{\circ}\text{C}}$     | $\sigma_{60^{\circ}\text{C}}$    | $T_0$      | $A^*$                            | $B$         | $R^2$ |
|--------|------------------------|-----------------------------------|----------------------------------|------------|----------------------------------|-------------|-------|
|        | $\text{O}/\text{Li}^+$ | $\text{Scm}^{-1}$                 | $\text{Scm}^{-1}$                | $\text{K}$ | $\text{Scm}^{-1} \text{K}^{1/2}$ | $\text{eV}$ |       |
| BEG1   | 4                      | $(4.37 \pm 0.89) \times 10^{-10}$ | $(1.52 \pm 0.90) \times 10^{-7}$ | 238        | 2.017                            | 0.0367      | 0.984 |
| BEG2   | 6                      | $(9.97 \pm 0.03) \times 10^{-9}$  | $(4.81 \pm 0.02) \times 10^{-7}$ | 234        | 2.971                            | 0.0333      | 0.996 |
| BEG4   | 10                     | $(6 \pm 0.04) \times 10^{-7}$     | $(4.96 \pm 0.01) \times 10^{-6}$ | 209        | 2.726                            | 0.0310      | 0.998 |
| BEG6   | 14                     | $(2.65 \pm 0.04) \times 10^{-6}$  | $(1.14 \pm 0.08) \times 10^{-5}$ | 189        | 2.915                            | 0.0275      | 0.998 |
| BEG8   | 20                     | $(4.95 \pm 0.02) \times 10^{-6}$  | $(1.46 \pm 0.04) \times 10^{-5}$ | 181        | 4.15                             | 0.0245      | 0.993 |

Figure 7c shows the pseudo-activation energy values as a function of the  $\text{O}/\text{Li}^+$  molar ratio values, where the lowest  $\text{O}/\text{Li}^+$  molar ratio (*BEG1*) presents the highest pseudo-activation energy  $B \sim 0.052 \text{ eV}$ , this is similar to the lowest values of  $B$  obtained for SLICPE by Zhu et al..<sup>40</sup> and it decreases with respect to the increase in the PEG chains length between  $sp^3$  boron atoms, until reaching an asymptotic behavior, where the pseudo-activation energies ( $B$ ) of  $\sim 0.025$  and  $\sim 0.023 \text{ eV}$  are obtained for *BEG6* and *BEG8*, respectively. To the best of our knowledge, these two  $B$  values are among the lowest for SLICPE.<sup>10, 35, 58</sup> On the other hand, it has been demonstrated that in the absence of organic solvents, the oxygen atoms of the PEG chains provide the necessary sites to transport  $\text{Li}^+$  through the polymeric matrix.<sup>59</sup>

This result, in addition to those obtained by DSC, shows that variation of the ionic conductivity and the pseudo-activation energy of the conductivity with  $\text{Li}^+$  concentration, are opposite to the expected ones. This indicates that the chains mobility and the dissociation of the  $\text{Li}^+$  and  $\text{BO}_4^-$  complexes play an important role in the predominant transport process in synthesized SLICPE. This, is particularly significant, in samples with the highest  $\text{O}/\text{Li}^+$  molar ratio values, where the



distance between boron atoms is the shortest and the polymer chains mobility is thus limited. This result suggests that for *BEG1* and *BEG2*, the ionic conduction mechanism is via  $Li^+$  hopping through oxygen sites of the  $\sim BO_4^- - CH_2CH_2 - BO_4^- \sim$  and  $\sim BO_4^- - CH_2CH_2EO - BO_4^- \sim$  chains, whereas for *BEG4*, *BEG6* and *BEG8*, the preferential ionic conduction mechanism via  $Li^+$  hopping through oxygen sites of the  $\sim BO_4^- - CH_2CH_2(EO)_n - BO_4^- \sim$  chains, but favored by the mobility of the chains associated to the PEG chain length between  $sp^3$  boron atoms.

The  $t_{Li^+}$  values were evaluated by AC-DC polarization experiments on Li0/SLICPE/Li0 cells. Figure 7d shows the AC-DC curves obtained for *BEG8* (The AC-DC curves for the other SLICPE are reported in Figure S1, Supporting Information). The current declines gradually over time (a typical time dependence response of DC polarization for *BEG8*, Figure 7d), and a slight cell resistance change ( $R_0 + R_s$ ) due to the passivation layer after the CA test is evaluated from impedance spectrum (inset of Figure 7d). The  $t_{Li^+}$  was evaluated using (Eq. (2)) and their values are reported in Table 3, which are between  $\sim 0.88$  and  $\sim 0.97$  at 25 °C, suggesting that the synthesized polymer electrolytes exhibit single ion conduction, within the limits of the implemented analytical technique. The variation of pseudo-activation energy of total conductivity and  $t_{Li^+}$  of synthesized SLICPE with the  $O/Li^+$  ratio is shown in Figure 7d and agree well with the above described change in the  $Li^+$  conduction mechanism. The monotonic behavior of both curves suggests that the low mobility of the polymeric matrix limits the  $Li^+$  transport in *BEG1*, generating high values of pseudo-activation energy and  $t_{Li^+}$ , exhibiting the typical behavior of ceramic electrolytes.<sup>60</sup>

TABLE 3. Measured values for the corresponding calculated values of lithium ion transference numbers ( $t_{Li^+}$ ) by Eq. 2.

| Sample | $I_0(\mu A)$ | $I_s(\mu A)$ | $R_0(\Omega)$ | $R_s(\Omega)$ | $t_{Li^+}$                        |
|--------|--------------|--------------|---------------|---------------|-----------------------------------|
| BEG1   | 0.05         | 0.015        | 5430900       | 5593827       | $(0.971 \pm 2.80) \times 10^{-3}$ |
| BEG2   | 0.08         | 0.015        | 2890600       | 3058254       | $(0.946 \pm 1.95) \times 10^{-3}$ |
| BEG4   | 0.12         | 0.015        | 912660        | 996624        | $(0.919 \pm 1.40) \times 10^{-3}$ |
| BEG6   | 0.17         | 0.045        | 773190        | 865972        | $(0.893 \pm 1.25) \times 10^{-3}$ |
| BEG8   | 0.22         | 0.02         | 387100        | 437423        | $(0.883 \pm 1.15) \times 10^{-3}$ |

The electrochemical stability of synthesized SLICPE was analyzed by LSV at 60 °C in the range of  $OCP$  to 6 V vs  $Li^0/Li^+$ , with a scanning rate of 10 mV s<sup>-1</sup> (Figure 8). In the case of *BEG2* (Figure 8i), the SLICPE membrane exhibits an electrochemical stability window of 5.4 V vs  $Li^0/Li^+$ . The electrochemical stability window decreases as the PEG chains length increases, 4.5, 4.3 and 4.2 V vs  $Li^0/Li^+$  for *BEG4*, *BEG6* and *BEG8*, respectively (Figure 8ii-iv). This diminution of electrochemical stability in SLICPE may be due to  $Li^+-BO_4^-$  pairs concentration decrease in the polymeric matrix and the increase of the  $BO_4^-$  groups mobility, thereby promoting a secondary reaction on the surface of the lithium electrodes.<sup>22</sup> *BEG4* exhibited a stable electrochemical window of at least 4.5 V vs  $Li^0/Li^+$ , which is similar to that of the PEO/lithium bis(oxalate)borate (LiBOB) composite electrolyte ( $\sim 4.5$  V vs  $Li^0/Li^+$ ).<sup>22, 44</sup>

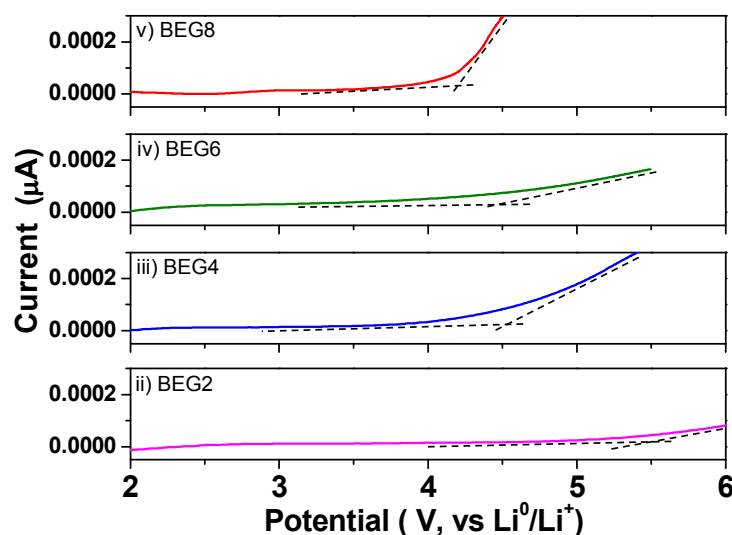


Figure 8. Linear sweep voltammograms ( $\nu=10 \text{ mV s}^{-1}$ ) obtained in the stainless steel/SLICPE/ $\text{Li}^0$  cell; ii) *BEG2*, iii) *BEG4*, iv) *BEG6* and v) *BEG8*.

The measure of the impedance on storage time for a symmetric  $\text{Li}/\text{BEG8}/\text{Li}$  cell has been performed (Figure S2, Supporting Information). The EIS spectra show a small modification between 8 and 10 days, indicating a possible BEG8 stabilized within 10 days, indicating good stability with lithium metal; this preliminary statement requires a longer time test.

Based on the best ionic conductivity values, sample BEG8 was selected for evaluation in a  $\text{Li}^0/\text{BEG8}/\text{LiFePO}_4$  cell. The cycling performance, at a current value of  $C/25$ , is shown in Figure S3, Supporting Information). Specific discharge capacity values, at  $30^\circ\text{C}$ , of  $\sim 30 \text{ mAhg}^{-1}$  and  $\sim 20 \text{ mAhg}^{-1}$  were obtained in the first and tenth cycles, respectively. This battery performance is limited by the electrolyte resistance and the poor interfacial contact on the electrode surface, which is evident at high potential values of the charge/discharge curves during cycling.

## Conclusions

In the context of developing superior SLICPE candidates compatible with cathodic materials for LIB with relatively high redox potentials ( $\sim 5$  V vs  $Li^0/Li^+$ ), a series of SLICPE based on  $sp^3$  boron and poly(ethylene glycol) bridges were synthesized. The series was synthesized based on a relatively simple and low cost route: a polycondensation reaction with  $LiB(OCH_3)_4$ , where the PEG chain length was varied from one to eight ethoxy groups.

The results of the characterization of the SLICPE by  $^1H$ ,  $^{13}C$ ,  $^{11}B$  and  $^7Li$  NMR, XRD, DSC, revealed significant structural changes as a function of the PEG chain length between borate groups: semicrystalline phases were present in samples with short PEG chains, while completely amorphous phases were found for the samples with the longest PEG chains. The latter furthermore presented the highest free volume values and the smallest rigidity (based on  $T_g$  values). Moreover, the results of the electrochemical techniques revealed a change in the governing ionic conduction mechanism as a function of chain length:  $Li^+$  hopping through oxygen sites of the ethoxy groups was the predominant transport mechanism in SLICPE with the shortest PEG chains, while an ionic motion promoted by the chain motion described the transport in the SLICPE with the longest PEG chains. The ionic conductivity varied accordingly to the described scenario and the pseudo-activation energy consequently decreased as the conduction mechanism changed. However, as the PEG chain length increases, the  $BO_4^-$  groups, as part of the polymeric matrix, become mobile (also as the polymeric matrix is less rigid), so that the  $t_{Li^+}$  is slightly decreased. Additionally, a slight decrease in the electrochemical stability was observed as the PEG chain length increases, as a result of the mobility of the  $BO_4^-$  groups. Thus, it was possible to obtain a relatively high  $t_{Li^+}$  as well as modulate the  $Li^+$  mobility and predominant conduction mechanism through the structural variation of samples caused by the effect of the PEG chain length.

## ASSOCIATED CONTENT

**Supporting Information.** S1. Lithium transfer number ( $t_{Li^+}$ ) evaluation. S2. LIB storage evaluation

## AUTHOR INFORMATION

### Corresponding Author

\* Tel.: +521- 5804- 4671 Ext. 12. E-mail igm@xanum.uam.mx

### Notes

The authors declare no competing financial interest.

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